

Introduction to LaTeX

Your Name

April 14, 2026

Abstract

This document demonstrates the basic features of $\text{\LaTeX}$ typesetting. It covers sections, mathematics, figures, and references.

Contents

1	Introduction	1
1.1	Why LaTeX?	1
2	Mathematics	1
2.1	Inline Mathematics	2
2.2	Display Mathematics	2
3	Conclusion	2

1 Introduction

$\text{\LaTeX}$ is a document preparation system for high-quality typesetting. It is widely used in academia for the communication and publication of scientific documents in many fields, including mathematics, computer science, engineering, physics, and economics.

1.1 Why LaTeX?

$\text{\LaTeX}$ provides several advantages over word processors:

- Professional-quality typesetting
- Excellent handling of mathematical notation
- Automatic numbering and cross-referencing
- Separation of content from formatting

2 Mathematics

One of $\text{\LaTeX}$'s greatest strengths is typesetting mathematics.

2.1 Inline Mathematics

Einstein's famous equation $E = mc^2$ changed physics forever. The quadratic formula is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

2.2 Display Mathematics

The Euler identity:

$$e^{i\pi} + 1 = 0$$

A more complex equation:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \tag{1}$$

See Equation 1 for the Gaussian integral.

3 Conclusion

L^AT_EX is a powerful tool for creating professional documents. This template can serve as a starting point for your own work.